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Commit decision

Hypothesis HC-01 · Email Thread Collapse

causal root candidate

SIM

S1 ingestion truncates email threads to the last 1–3 messages when total message count ≥ 8 . The resulting context loss propagates downstream and surfaces as C2 faithfulness failures (tense_collapse, unsourced_claim) and degrades the Audio-Native energy arc. Predicted: thread length is the dominant driver of C1+C2 failure, not retriever or generator quality.

Source: eval-2847 / -2869 / -2811 + Surface Mapper Sample: n=186 release-gate cases Owner: D. Chen · Eval Started: 6h ago Linked tag: C1.ingestion.email.thread_collapse

Cohort cuts — failure rate by candidate driver all 186 cases · C1+C2 joint failure + Add cut

COHORT	N	C1+C2 FAIL	BAR	VS BASELINE	GATE
Baseline · all 186 cases	186	38%		—	ref
Thread length ≥ 8 KEY CUT	62	78%		+2.1×	X FAIL
Thread length 1–7 (control)	124	18%		-0.5×	✓ pass
Profile A · lawyer	74	52%		+1.4×	—
Profile B · founder	62	33%		-0.1×	—
Profile C · researcher	50	22%		-0.6×	—
Retriever recall@10 < 0.6	41	51%		+1.3×	—
Generator temp ≥ 0.7	29	45%		+1.2×	—
Attachment present	38	39%		+1.0×	—

Driver weights — logistic regression, C1+C2 fail outcome · 95% CI

thread_len_ge_8

SIM

+2.41

p<0.001

profile_a_lawyer

+0.92

p=0.02

retriever_recall_low

+0.78

p=0.04

generator_temp_high

+0.41

p=0.18 ns

attachment_present

+0.08

p=0.61 ns

thread_recency_days

-0.22

p=0.27 ns

Model fit: McFadden R² 0.31 · n=186 · target C1+C2 fail. thread_len_ge_8 dominates; retriever and profile contribute independently but at ~⅓ the magnitude. Two non-significant drivers retained for transparency.

Confound check — does retriever miss explain the S1 signal away?

thread_len ≥ 8 unadjusted

+2.41

thread_len ≥ 8 adj. for retriever recall

+2.18

thread_len ≥ 8 adj. for profile + temp

+2.04

Coefficient survives adjustment: 15% attenuation, not collapse. Retriever miss is correlated but not the cause — it's another *symptom* of upstream truncation.

Counter-factual probe — 24 cases re-run with full thread

PRODUCTION S1 (TRUNCATED)

79%

C1+C2 fail · 19/24

SYNTHETIC FULL-THREAD S1

21%

C1+C2 fail · 5/24

$\Delta -58$ pts

when threads not truncated · paired-test p < 0.001

SIM

Holding model, retriever, profile and prompt constant — only S1 behavior changed. The 5 residual failures are independent: 3× retriever miss, 2× generator hallucination on full context.

Causal chain — what we now believe

S1 · RETRIEVER

REV

Thread collapse

thread_len ≥ 8

→

S6 · SURFACE

SIM

Tense collapse

unsourced_claim

→

S9 · OUTCOME

User trust

energy_de

Retriever miss (S4) was an alternative root candidate — confound check ruled it out as a correlated symptom. Generator temperature is independent and small.

Decision gates — per T-13 protocol, ≥ 3 of 4 must pass to commit root

✓ PASS

61

Does cohort cut yield $\geq 2\times$ lift?

thread_len $\geq 8 \rightarrow +2.1\times$ baseline · cohort n=62 sufficient.

✓ PASS

62

Does coefficient survive confound adjustment?

15% attenuation after adjusting for retriever + profile + temp. Not collapsed.

✓ PASS

63

Does counter-factual remove the failure?

Synthetic full-thread S1: -58 pts failure rate · paired p<0.001.

X FAIL

64

Is evidence LIVE-grade?

All evidence is SIM / REV. No LIVE path yet — needs prod sampler before final commit.

3 OF 4 PASSED — commit allowed with SIM-grade caveat. LIVE confirmation queued as T-15 experiment.

Recommended actions — route into Backlog T-17 / Brief T-01 / Experiments T-15

1

Commit HC-01 as confirmed causal root (SIM-grade)

Mark C1.ingestion.email.thread_collapse as root for HC-01 in Causal Chains. Auto-tag retriever recall + generator temp as correlated symptoms.

COMMIT

2

Open backlog item: lift S1 thread window to full thread for length ≤ 50

Owner: Ingestion eng · expected -15 pts release-gate failure if shipped this sprint · risk: latency +120ms p95.

+ T-17

3

Promote evidence floor SIM $\rightarrow$ LIVE for thread-length driver

1% prod sampler on email-thread traces · 200 cases · 5d. Required by G4 before final root commit.

+ T-15 EXPERIMENT

4

Update T-01 Brief: add thread_len_ge_8 stratification to release-gate sampler

Current sampler under-represents long-thread cases by 2×. Will surface this class earlier on the next run.

$\rightarrow$ T-01

5

Close HC-01b "retriever miss is the root" — superseded

Confound-adjusted coefficient ruled retriever recall a correlated symptom, not cause. Retain as monitored variable, not driver.

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